



PES University, Bengaluru
(Established under Karnataka Act 16 of 2013)

END SEMESTER ASSESSMENT (ESA) - DEC 2023

UE20CS204 - Web Technologies

Total Marks : 100.0

1.a. What is HTTP? Explain the structure of HTTP request message.
(4.0 Marks)

1.b. With a neat diagram, explain the CSS Box model and its significance.
(4.0 Marks)

1.c. Write Code to get following output using lists and table

(6.0 Marks)

list to be made

1. egg
2. meat
3. milk
- veg

• non veg

my kichen

my home

egg

good for health

Books Details

book_name	year
java	2009
c++	2019

1.d. Write code to get following output

Fill your details

Personal Info

Name

LastName

Enter your email id

Password

Enter your address

Select Gender

Gender

Select Gender

☐ Male

☐ Female

Select fav lang

Fav lan

☐ HTML

☐ C++

☐ Java

Submit

Reset

(6.0 Marks)

2.a. Write Breifly about the two methods of the geolocation object
(4.0 Marks)

2.b. What is web worker?

Name the methods/events supported by web workers for communication between worker and main threads. Show how they are used with an example. (4.0 Marks)

2.c. Write a script to display below texts in web page and onclick on **Call of Duty** and **Assassins Creed** texts the colour of the text should change to red.

Popular Games

- Call of Duty
- Assassins Creed
- PUBG

(6.0 Marks)

2.d. Write a code to get the following output in browser using html canvas (6.0 Marks)



3.a. Write a code to get the following output in react using components, Create a class component called Article1 which is having states such as name,color and price
if user clicks change color button text red should get changed blue.

my car color is red

price is 10000000

Change color

(6.0 Marks)

3.b. What is the significance of key property? Describe with an example.
(4.0 Marks)

3.c. Create a class component which will render a H1 and paragraph element apply different inline style for H1 and P element created using java script objects (key value pair).
(6.0 Marks)

3.d. With a neat diagram explain any 4 component in component life cycle. (4.0 Marks)

4.a. What are buffers in Node JS? Write a program to create a buffer, write data to buffer and read data from buffer (4.0 Marks)

4.b. In a MongoDB database, there is a "course" collection. From the collection, write queries to do the following:

- List all documents
- List documents with code = "UE20CS204"
- List the first document with credits = "4"

(3.0 Marks)

4.c. Write a code in node .js server to

1. Write data into file called input.txt

2. Read data Asynchronously from input.txt
handle the error properly

(6.0 Marks)

4.d. Write code snippet to connect to a db named "Groceries" on a MongoDB instance running on localhost:27017 and write the following details into the "Fruits" collection, insert many documents (at least 5) with details: Fruit-name and price.

It should handle errors appropriately.

Note: write only the code required to write into the database. (7.0 Marks)

5.a. What is RESTful API? Explain any 4 design specification/constraints of REST API.

(4.0 Marks)

5.b. Write a code to collect form data for the library card which has fields name,sem,branch and Submit in pug template and include pug template in express code to retrieve the form data. (8.0 Marks)

5.c. MBappe, a back end developer, has to create at least 4 -5 Express routes each for the resources "players", "match" and "teams". Since he is new to Express, he wants suggestions on how to structure his JavaScript code. Write an approach that can be taken by him to solve this problem. (4.0 Marks)

5.d. Write a code in express to create a server which have 2 different route paths like /pes/:branch,/pes/branch/id which returns 3 different contents when particular route has been entered in browser if user enters following paths server should return following contents

/pes/:CSE—> branch name is CSE
/PES/:CSE/:123→id:123 and name :CSE (4.0 Marks)